

MYS BILLINGS, Russian FederationWMO: **250620**Lat: **69.877N**Lon: **175.755E**Elev: **3**StdP: **101.29**Time Zone: **12.00 (E12)**Period: **09-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-39.2	-37.4	-42.5	0.1	-39.0	-40.6	0.1	-37.3	15.3	-13.1	13.4	-20.3	2.7	230	1.087

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.0	15.8	12.3	12.7	10.2	9.8	8.1	12.3	15.7	10.2	12.8	8.1	9.8	5.7	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.5	7.9	14.2	8.1	6.7	12.0	6.4	6.0	9.7	35.1	15.8	29.8	12.8	25.1	9.9	23.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.1	12.5	11.3	DB	-42.4	21.6	2.8	3.2	-44.4	23.9	-46.1	25.8	-47.7	27.7	-49.7	30.0	(4)
(5)			WB	-41.6	16.6	2.7	3.4	-43.6	19.0	-45.2	20.9	-46.7	22.8	-48.7	25.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-9.8	-24.5	-24.3	-23.4	-15.5	-5.3	2.1	4.6	4.0	2.0	-4.7	-12.8	-20.2
	DBStd	12.56	6.98	7.81	7.37	6.49	4.62	3.15	3.21	3.07	3.08	5.21	6.88	8.52
	HDD10.0	7230	1069	960	1037	766	475	240	175	191	242	455	685	936
	HDD18.3	10258	1327	1193	1295	1016	733	488	427	446	491	713	935	1194
	CDD10.0	14	0	0	0	0	0	2	7	3	2	0	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	4	0	0	0	0	0	0	4	0	0	0	0	0
	CDH26.7	1	0	0	0	0	0	0	1	0	0	0	0	0
Wind	WSAvg	5.1	4.9	4.6	4.7	4.8	4.8	5.4	5.4	5.7	5.5	5.3	5.3	4.9
Precipitation	PrecAvg	454	32	35	24	22	27	29	47	51	50	57	46	34
	PrecMax	656	93	87	57	43	59	53	96	93	133	177	91	79
	PrecMin	302	4	3	1	9	5	12	12	19	12	12	10	5
	PrecStd	84	17	20	15	9	13	12	21	21	25	31	18	20
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.2	-0.7	0.5	1.5	6.8	18.0	22.2	17.6	15.3	5.2	0.8	-0.7
		MCWB	-1.3	-1.8	-0.7	0.3	5.2	12.4	16.6	13.6	11.4	3.9	-0.3	-1.8
	2%	DB	-4.8	-6.7	-3.2	-2.3	2.9	14.0	16.9	13.9	11.5	3.0	-0.3	-3.8
		MCWB	-5.4	-7.3	-4.1	-3.0	1.6	10.1	13.6	11.1	9.4	2.5	-0.9	-4.5
	5%	DB	-10.9	-10.4	-8.9	-4.2	1.4	9.9	12.5	10.7	8.1	2.1	-1.7	-6.7
		MCWB	-11.3	-10.7	-9.4	-4.7	0.5	7.9	10.4	8.9	6.8	1.6	-2.2	-7.2
	10%	DB	-15.8	-14.2	-12.8	-6.2	0.4	6.7	9.1	8.0	5.0	1.2	-3.5	-8.7
		MCWB	-16.2	-14.5	-13.2	-6.6	-0.1	5.2	7.7	6.8	4.1	0.6	-4.1	-9.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.2	-1.9	-0.5	0.3	5.0	12.6	17.5	14.3	11.7	4.0	-0.1	-1.8
		MCDB	-0.2	-0.6	0.5	1.5	6.7	18.0	20.9	16.9	14.9	5.0	0.2	-0.9
	2%	WB	-5.2	-7.4	-4.1	-2.9	1.7	10.3	13.3	11.3	9.3	2.6	-0.9	-4.4
		MCDB	-4.5	-6.8	-3.3	-2.3	2.5	13.9	16.6	13.8	11.3	2.9	-0.4	-3.9
	5%	WB	-11.3	-10.8	-9.4	-4.7	0.6	7.9	10.5	9.0	6.5	1.7	-2.2	-7.2
		MCDB	-11.0	-10.4	-8.9	-4.2	1.2	10.2	12.7	10.8	8.1	2.0	-1.7	-6.7
	10%	WB	-16.1	-14.5	-13.2	-6.5	-0.1	5.2	7.7	6.8	4.3	0.8	-4.1	-9.2
		MCDB	-15.9	-14.3	-12.8	-6.1	0.3	6.6	9.1	7.9	4.9	1.2	-3.6	-8.7
Mean Daily Temperature Range	5% DB	MDBR	6.1	6.1	6.4	7.0	4.9	4.3	5.0	3.6	2.9	3.9	6.1	5.7
		MCDBR	11.0	9.7	9.6	7.0	5.0	10.5	12.1	9.8	7.7	2.7	5.3	7.7
		MCWBR	10.6	9.3	9.1	6.7	4.3	7.8	9.6	7.7	5.9	2.4	5.2	7.5
	5% WB	MCDBR	10.9	9.4	9.6	7.1	4.5	10.6	12.1	9.8	7.4	2.6	5.2	7.6
		MCWBR	10.6	9.0	9.2	6.8	3.9	7.9	9.8	7.8	5.8	2.3	5.2	7.4
Clear-Sky Solar Irradiance	taub	N/A	0.208	0.281	0.327	0.354	0.352	0.373	0.377	0.312	0.248	N/A	N/A	
	taud	N/A	2.025	2.032	1.960	2.012	2.232	2.311	2.313	2.449	2.121	N/A	N/A	
	Ebn at Noon	0	535	704	782	811	834	800	742	708	454	0	0	
	Edh at Noon	0	40	85	127	137	114	103	91	59	36	0	0	
All-Sky Solar Radiation	RadAvg	0.00	0.34	1.98	4.32	5.96	6.11	4.97	3.18	1.63	0.48	0.00	0.00	
	RadStd	0.00	0.04	0.13	0.20	0.34	0.41	0.44	0.31	0.13	0.06	0.00	0.00	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page